Duality AI’s Offroad Semantic Scene Segmentation

Team Name: ZeroDay

Project Title: Duality AI’s Offroad Semantic Scene Segmentation

Team Members:

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Project Summary

Semantic segmentation for complex off-road environments using DINOv2 transformer backbone.

Backbone frozen; custom convolutional decoder trained for 10 semantic classes.

Motivation & Real-World Relevance

- Autonomous vehicles and robots
- Search and rescue operations
- Defense and surveillance systems
- Agricultural and mining automation

Dataset Description Dataset Name:
Offroad_Segmentation_Training_Dataset Dataset Structure includes train/val splits with Color_Images and Segmentation folders.
Number of Classes: 10 semantic classes
Raw pixel values mapped to class indices using custom mapping strategy.

Image Preprocessing Resolution:

266 × 476 pixels (divisible by 14 for DINOv2 compatibility).
Images resized using Bilinear interpolation. Masks resized using Nearest-neighbor interpolation.
Normalization using ImageNet mean [0.485, 0.456, 0.406] and std [0.229, 0.224, 0.225].

Model & Training Configuration

Backbone: DINOv2 ViT-S/14 (self-supervised pretrained).

Loss: CrossEntropy + 0.5 × Dice Loss.

Optimizer: SGD (momentum 0.9).

Learning Rate: 1e-4.

Batch Size: 2.

Epochs: 30.

Results & Performance Metrics

Metric	Value
Best Epoch	29
Final Validation IoU	0.4952
Validation Dice Score	0.65
mAP@50	0.65
Validation Pixel Accuracy	0.80

Best validation performance achieved at Epoch 29. Transformer-based global feature extraction combined with Dice-enhanced optimization improved region overlap and robustness.

Challenges & Future Work

Class imbalance and occlusions addressed using Dice loss.

Future: Backbone fine-tuning, multi-scale decoder, real-time optimization.

Final Confirmation

Original work completed by Team ZeroDay.

Test images were not used in training/validation.

All required submission documents verified.
